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RESEARCH ARTICLE

Sankalp: AI-Powered Career Guidance System

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ABSTRACT Traditional career guidance systems often lack personalization, adaptability, and accessibility, leading to suboptimal decisions in a rapidly changing job market. This study introduces **Sankalp**, an AI-driven career guidance framework that combines semantic reasoning, emotion analysis, and adaptive learning to provide inclusive and explainable recommendations. Built on a multi-agent hybrid architecture, Sankalp integrates modules for semantic matching using Sentence-BERT (SBERT), emotion-aware feedback through Valence Aware Dictionary and Sentiment Reasoner (VADER) sentiment analysis, a Naukri API–based job trends agent, a knowledge graph, and a voice assistant that supports English, Hindi, and Kannada. A reinforcement learning–based fusion layer dynamically refines recommendations based on real-time user feedback, modeling career choice as a life-span adaptive process. It was evaluated against a Curated Ground Truth (CGT) dataset using 160 participant profiles, Sankalp demonstrated superior performance, achieving a Top-1 Hit Rate of 77.5%, a Top-3 Hit Rate of 90.5%, and a high mean user satisfaction score of 4.71/5.0. The latency analysis confirmed high operational efficiency, with an average end-to-end response time of 1.25 s. Through its inclusive design and adaptive, explainable hybrid AI, Sankalp enhances the accessibility, reliability, and ethical delivery of career guidance services, aligning with the global sustainable development goals.

INDEX TERMS AI-driven career guidance, emotion analysis, google cloud speech-to-text, knowledge graph, multi-agent architecture, multilingual support, natural language processing, personalized recommendations, rule-based AI, Sankalp, sentence-BERT, sustainable development goals, VADER sentiment analysis, voice assistant.

I. INTRODUCTION

In the increasingly dynamic global landscape, the importance of effective career guidance cannot be overstated. For individuals, well-informed career decisions are essential for personal fulfillment, economic stability, and societal contribution [25]. However, a lack of clear direction can lead to underemployment, skill mismatches, and reduced productivity, affecting both individual well-being and national economic growth. This challenge is particularly pronounced in developing nations with large youth populations, where personal guidance is significantly lacking. For example, India faces a severe disparity with a student-to-counselor ratio of approximately 1:3,000, which is dramatically below the globally recommended standard of 1:250 [25], [26]. This shortage leaves most students without adequate support to navigate an increasingly complex job market in the future. This paper introduces Sankalp, an innovative AI-driven system designed to democratize access to personalized and dynamic career counseling, thus bridging the critical gap between educational outcomes and employment opportunities.

India, with its vast and diverse youth demographic, is at a critical juncture. The 2020 National Education Policy (NEP) envisions a transformative shift in the country's education system, emphasizing holistic development, skill enhancement, and a stronger link between learning outcomes and employment opportunities [13]. A core tenet of NEP 2020 is the imperative for comprehensive career guidance, recognizing its pivotal role in enabling students to navigate diverse learning pathways and make informed choices aligned with their aptitudes and aspirations. The policy explicitly calls for integrating career counseling at all stages of schooling, with the aim of preparing a future-ready workforce and positioning India as a global knowledge

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superpower. This policy framework underscores the urgent need for scalable, accessible, and effective career guidance solutions that can meet the unique demands of India's multifaceted educational landscape [12].

Despite the recognized importance of career guidance, significant barriers persist in its widespread and equitable delivery. Firstly, access remains a critical issue. Traditional career counseling services are often concentrated in urban centers, leaving a large majority of students in rural or semi-urban areas with limited or no access to professional guidance [4]. This creates a stark urban-rural divide in opportunity. Secondly, there is a severe shortage of trained career counselors. The sheer scale of the student population in India greatly outweighs the availability of qualified professionals, leading to high student-to-counselor ratios that compromise the quality and personalization of the guidance [5]. Third, language presents a substantial barrier. India's rich linguistic diversity means that guidance materials and interactions often need to be available in multiple regional languages, a challenge that most conventional systems struggle to meet, thus excluding a significant portion of the student population [7]. Finally, many students face a lack of motivation or feel overwhelmed by the multitude of career options, often due to insufficient information, social pressures, or a disconnect between their academic pursuits and the relevance of the real-world career [9]. These cumulative barriers prevent students from reaching their full potential and making optimal career choices.

To address these multifaceted challenges, this paper proposes **Sankalp**, an innovative AI-driven career guidance system designed to provide personalized, accessible and dynamic career recommendations. Sankalp's foundational methodology is rooted in established psychological models of vocational choice, specifically the person-environment fit theory [27]. Sankalp distinguishes itself through a sophisticated *multi-agent architecture*, where specialized modules collaborate to provide comprehensive guidance [6], [7]. Its core intelligence integrates *Emotion-aware Natural Language Processing (NLP)*, utilizing **VADER sentiment analysis** [18] to understand the user's emotional state during interaction, allowing the system to respond with greater empathy and adapt its conversational flow for a more supportive experience. The system offers seamless voice and text input/output, leveraging **Google Cloud Speech-to-Text** and **Text-to-Speech** services, coupled with **Google Cloud Translation**, to provide truly *multilingual interaction* in *English, Hindi, and Kannada* [4], [9]. This multimodal and multilingual capability significantly improves accessibility and user engagement.

A novel component of Sankalp is its **rule-based Reasoning Engine**, which provides logical, interpretable filtering and prioritization of career recommendations. This rule-based structure is informed by established methodologies for structured NLP query processing [31]. To overcome the inherent rigidity of purely rule-based systems, this engine is enhanced with an adaptive Reinforcement Learning (RL) layer to dynamically refine recommendation weights based on user feedback. The RL layer, along with the system's capacity to evolve recommendations, reflects career development as a life-span process that evolves with experience and self-concept [28]. The rule engine encodes domain-specific eligibility constraints (for example, academic thresholds) to ensure transparency and explainability [2]. This hybrid system works in tandem with advanced AI modules to create an explainable and adaptable recommendation framework [3].

Furthermore, Sankalp incorporates a real-time job market trend module that uses the Naukri API to ensure that recommendations align with current industry demands and future prospects [6]. The system also provides graphical explainability through its integrated **Knowledge Graph**, visually representing the interconnections between careers, skills, and subjects, which justify recommendations and empower users with a clearer understanding of their career paths [10], [11].

Key Contributions: The development of Sankalp represents several significant contributions to the field of AI-driven career guidance. Primarily, it introduces a modular, multimodal, and multilingual career recommendation system that integrates robust market alignment capabilities (via the Naukri API), ensuring relevance and practicality [5]. A novel aspect is the hybrid recommendation model, which intelligently combines rule-based reasoning, an adaptive Reinforcement Learning layer [28], and advanced semantic similarity techniques, notably **Sentence Transformers (SBERT)** [17]. The semantic matching module uses verbal analysis to accurately interpret user intent [30]. Furthermore, the system operationalizes the principles of social cognitive career theory (SCCT) by factoring in user confidence and self-efficacy through its adaptive feedback loops [29]. Sankalp is designed with explainability in mind, providing visual graphs through its integrated Knowledge Graph to justify recommendations and clearly illustrate the interconnected paths between various careers [10]. The effectiveness of the system has been empirically demonstrated by evaluating against a Curated Ground Truth (CGT), which yielded a $\mathbf{Top-3}$ Hit Rate of 90.5% (95% CI: [88.1%, 92.9%]), along with a high mean user satisfaction score of 4.71/5.0. Ultimately, Sankalp pioneers a scalable AI solution that directly addresses critical barriers to career guidance in various linguistic and socioeconomic contexts, thereby seamlessly aligning with the objectives of NEP 2020 and the Sustainable Development Goals of the United Nations [14], [15].

The remainder of this paper is organized as follows. Section II presents the Literature Survey. Section III describes the Proposed Sankalp System Architecture, followed by System Implementation in Section IV. The Experimental Setup is detailed in Section V, and the Results and Analysis are presented in Section VI. Section VII provides a Discussion, while Section VIII outlines Future Work. Section IX

highlights Reproducibility aspects, and Section X discusses Ethics and Responsible AI considerations. Finally, Section XI concludes the paper.

II. LITERATURE SURVEY

The landscape of career guidance has undergone a significant transformation with the advent of artificial intelligence, moving beyond traditional counseling to embrace personalized approaches and data-driven. This section critically reviews existing research and prominent online platforms, highlighting their contributions and limitations, thus establishing the unique position and comprehensive novelty of the Sankalp Career Guidance System.

A. REVIEW OF ACADEMIC SYSTEMS

Recent academic endeavors have explored various aspects of AI in career guidance, each contributing valuable insights but often lacking the comprehensive, multimodal, and multilingual integration that Sankalp offers.

"AI in education: A Review of Personalized Learning and Educational Technology" by Ayeni et al. [1] provides a wider contextual understanding of the role of AI in education. Although this paper does not propose a specific system, it highlights the increasing need for personalized learning and addresses critical challenges such as the digital divide, algorithmic biases, and ethical considerations in AI adoption within educational technology. Sankalp directly contributes to the advancements discussed in this review by providing a practical, scalable, and inclusive AI-driven solution. Its multimodal and multilingual design directly mitigates the digital and linguistic divides, offering personalized guidance that aligns with the review's call for equitable and effective educational technology, while its focus on semantic matching aims to reduce biases inherent in simpler recommendation models.

Jadhav et al. [2] proposed an NLP-powered data-driven career decision methodology implemented as a chatbot application, specifically targeting Indian students who have completed their 10th or 12th grade. Their system focuses on individualized counseling, utilizing a Neural Network Model for intent recognition and providing updates and course information. Although highly relevant to the Indian context, their system is fundamentally text-based and does not explicitly detail comprehensive multilingual support for conversational interaction. Sankalp differentiates itself by offering integrated voice input/output and robust multilingual capabilities (English, Hindi, Kannada), making it significantly more accessible across India's linguistic diversity. Sankalp's use of SBERT [17] for nuanced semantic matching and VADER sentiment analysis [18] for real-time emotional adaptation also provides a more sophisticated and empathetically responsive user experience.

Avinash et al. [3] proposed an AI-driven system based primarily on personalized recommendations, identifying skill gaps, and facilitating workforce planning by incorporating real-time job market data. Their methodology integrates Natural Language Processing (NLP) for profile analysis, Reinforcement Learning (RL) for optimizing recommendations, and Collaborative Filtering (CF) to identify trends. Although robust in its analytical capabilities, their work primarily operates within a text-based framework and does not explicitly detail the provision of multimodal input/output or comprehensive support for linguistically diverse users. Sankalp significantly distinguishes itself by providing a truly multimodal (voice and text) and multilingual (English, Hindi, Kannada) conversational interface, leveraging Google Cloud Speech-to-Text and Text-to-Speech services. This fundamental design choice markedly improves accessibility and user engagement for a broader and more linguistically diverse population. Furthermore, Sankalp's explicit use of Sentence Transformers (SBERT) [17] for nuanced semantic matching and VADER sentiment analysis [18] for empathetic interaction offers a more refined and responsive approach to understanding user profiles and dynamically adapting system responses.

Similarly, Birajdar et al. [4] introduced an AI-powered career counseling web application designed to deliver personalized recommendations, conduct skills assessments, and outline customized educational pathways. Their system analyzes individual student profiles and integrates with real-time job market data, emphasizing a user-friendly web interface and an interactive chatbot. However, their proposed system does not explicitly incorporate voice interaction as a primary modality and does not detail robust multilingual capabilities beyond implied English. Sankalp, in contrast, is fundamentally designed with integrated Google Cloud STT and TTS for seamless voice interaction, coupled with Google Cloud Translation for comprehensive multilingual support. This direct integration provides a more inclusive and natural user experience, crucial for widespread adoption in diverse regions.

Kulugh et al. [5] presented a personalized AI-based career guidance tool built on a theoretical foundation that merges Holland's Vocational Theory (HVT) and Social Cognitive Theory (SCT). Their system focuses on tailoring recommendations to individual aspirations through a structured development approach, utilizing Supabase for back-end services. While commendable in its theoretical grounding and systematic design, their system, like many others, lacks the explicit multimodal and multilingual features that are central to Sankalp's design. Sankalp's distinct advantage lies in its ability to engage users through both voice and text in multiple Indian languages, fostering a deeper connection and broader reach. Moreover, Sankalp's application of SBERT [17] for precise semantic matching and VADER sentiment analysis [18] for emotional intelligence provides a more granular and empathetically responsive understanding of user needs, surpassing the general AI layer described in their work.

The "VocaVisionary: A Career Guidance Chatbot" by Arshad et al. [6] offers a direct point of comparison, as it also utilizes generative AI (specifically Chat-Bison,

a fine-tuned PaLM 2 model) for conversational career guidance and includes an aptitude test module. However, VocaVisionary is primarily a text-based chatbot, with voice recognition noted as a future enhancement rather than an inherent capability, and no explicit mention of comprehensive multilingual support. Sankalp's inherent multimodal and multilingual design from its inception offers a significant leap in accessibility and immediate utility. Furthermore, Sankalp employs a hybrid AI approach [2], combining SBERT [17] for deep semantic personalization with VADER sentiment analysis [18] for empathetic interaction, which provides a more specialized and actionable guidance that goes beyond the general-purpose generative AI model's capabilities in this specific domain.

Integration of Vocational Psychology Theories: The psychological core of Sankalp integrates three seminal theories. Holland's Theory [27] establishes the trait-matching foundation using the six-type RIASEC model (Realistic, Investigative, Artistic, Social, Enterprising and Conventional) for personality-environment fit. Super's Life-Span Approach [28] guides the system's dynamic re-ranking and Reinforcement Learning (RL), reflecting the lifelong, evolving nature of career choice and self-concept. Social Cognitive Career Theory (SCCT) [29] ensures adaptability by incorporating user-reported factors such as self-efficacy and goal setting to suggest attainable and aspirational paths. The adaptive mechanisms of Sankalp are further informed by the foundational work on systems that learn and evolve [32].

Methodologies for Textual Interpretation: Sankalp employs established techniques for precise semantic and structured processing. Verb-Based Semantic Modeling [30] is the crucial methodology to interpret user language by identifying verbs as core indicators of intent, ensuring semantic precision in mapping conversational patterns to skills. This process is supported by Rule-Based Query Processing [31], which uses predefined linguistic rules and pattern matching logic to accurately process diverse user input during the early stages of analysis.

Comparison with Existing Hosted Platforms Beyond academic research, several online platforms currently offer career guidance services. Although many provide valuable resources and broad information, they often lack the depth of personalization, multimodal interaction, and comprehensive multilingual support that Sankalp provides.

Naukri Career Navigator [20] is a widely used platform in India that works as a job search engine with supplementary career advice and resume building tools. Its career recommendations often rely on keyword matching and broad categories. Although it offers extensive job listings, its ''market awareness'' is largely retrospective, based on existing vacancies rather than predictive trends. Crucially, it lacks an interactive conversational AI and does not integrate voice input/output or real-time multilingual support beyond basic textual content.

The IndiaSkills initiative and the National Career Service (NCS) [21] Portal are government-supported platforms in India aimed at skill development and career facilitation. The NCS portal provides job matching, counseling services (through registered human counselors), and information on skill development programs. Although the NCS portal offers content in multiple Indian languages, its interactive elements are primarily text-based, and it does not feature an AI-driven conversational agent with multimodal capabilities or real-time emotional intelligence. The depth of personalized recommendations, particularly those considering individual academic marks or personality traits, is also less sophisticated compared to Sankalp [7].

CareerAI (as a representative of general AI-driven career tools) [22] and MyNextMove (a prominent US-based platform powered by O*NET data) [23] offer career exploration tools. MyNextMove, for example, provides extensive occupational information, including required skills, educational pathways, and job outlooks. However, these platforms are predominantly text-based, primarily in English, and lack dynamic, real-time conversational AI with integrated emotion recognition. Their ''explainability'' is typically delivered through static, detailed descriptions rather than interactive, graphical representations of interconnected career pathways [8], [9]. A key limitation across these platforms is the absence of the seamless, integrated multimodal and multilingual conversational experience that is central to Sankalp's innovative design. The following Table 1 summarizes the key differentiators between Sankalp and a selection of existing tools and platforms, highlighting Sankalp's unique value proposition:

III. PROPOSED SANKALP SYSTEM ARCHITECTURE

The Sankalp Career Guidance System is engineered as a robust, modular, and intelligent platform meticulously designed to provide personalized, multimodal, and multilingual career guidance. Its architecture is carefully designed to ensure scalability, efficiency, and a rich, intuitive user experience. This section delineates the overall system architecture, followed by a comprehensive explanation of each core component and their intricate interconnections, providing every minute detail of their functionality in a formal academic tone.

A. OVERALL SYSTEM ARCHITECTURE

The Sankalp system operates on a client-server architecture, featuring a user front-end and a Flask-based back-end API. The API feeds requests to the Core AI Logic, which generates personalized roadmaps by coordinating various agents: Semantic Matcher, Emotion Analyzer, Sector Analyzer, Web Scraping Agent, and Knowledge Graph. Multimodal and multilingual support is handled by the Voice Assistant agent, using services from the Google Cloud Platform.

The architecture is divided into the following major layers as seen in Figure 1:

TABLE 1. Comparison of AI-powered career guidance systems.

Feature	CareerAI[22]	MyNextMove[23]	Naukri Career Navigator[20]	IndiaSkills / NCS Portal[21]	Sankalp (Proposed)
Language Support (Voice & Text)	English	English	English (primary)	Multiple Indian Languages (text only)	**English, Hindi,** Kannada (Voice & Text)
Input Modality	Text	Text	Text	Text	**Voice & Text**
Personalization Depth	Basic	Moderate	Basic	Moderate	**High (Interests,** Skills, Marks, Personality, Work Env.)
Market Awareness	No	Yes	Yes (Job Listings)	Yes	**Yes (** Real-time Scraping)
Emotion Awareness	No	No	No	No	**Yes (VADER)**
AI Explainability	No	Limited	Limited	No	**Yes (Graphical** Knowledge Graph)
Actionable Roadmap	No	Basic	Limited	Basic	**Detailed (Educational** Path, Course Links, Skills, Trends)
Feedback / Refinement Loop	No	No	No	No	**Explicit (Tracks** Rejected Careers)
Target Audience	General	General	Job Seekers	Students, Job Seekers	**Students & Young** Professionals (India-focused)
Offline Use	No	No	No	No	**Yes**

1) **User Experience Layer**: Developed using React.js, this layer provides an intuitive interface where students can either type their inputs or speak naturally using voice input. The UI supports real-time feedback and displays recommended career paths, required skills, and job trends.
2) **Application Integration Layer**: This layer orchestrates data flow between the front- and back-end agents. It handles input processing, calls emotion analyzers, performs semantic matching, applies decision rules, and returns the final recommendations.
3) **Cognitive Processing Layer (Multi-Agent System)**:
   - **Input Interpretation Module**: Converts voice to text using SpeechRecognition and preprocesses text.
   - **Emotion Analyzer**: Applies VADER sentiment analysis to extract emotional tone.
   - **Contextual Relevance Engine**: Uses SBERT embeddings to match user input with skill and career data.
   - **Deterministic Reasoning Engine**: Applies hand-crafted IF-THEN rules to recommend careers.
   - **Ontological Inference Engine**: Traverses subject-skill-career relationships to enrich results.
   - **Labour Market Intelligence Extractor**: Scrapes real-time job portals to add trending roles and demand data.
4) **Data Management Layer**: A PostgreSQL database stores career paths, skill mappings, job trends, and past user feedback to enable continuous learning and adaptation.

B. CORE AI LOGIC

The core AI Logic module represents the intelligent nucleus of Sankalp, meticulously integrating and orchestrating various specialized AI agents to provide comprehensive and personalized career guidance. This central component is responsible for managing the overall flow of information and decision making within the system ref Algorithm 1.

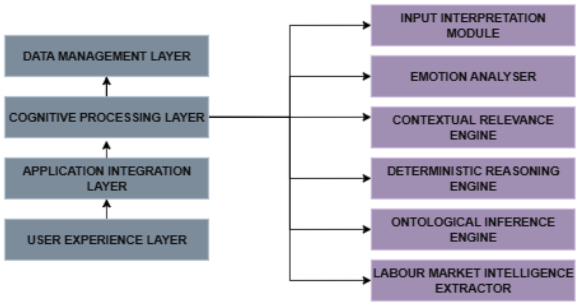


FIGURE 1. System architecture.

1) DATA MANAGEMENT

All foundational career-related information, which forms the bedrock of Sankalp's recommendations, is meticulously stored in a structured career data repository. This repository is designed as a comprehensive collection of various careers, where each individual career entry is enriched with a multitude of attributes. These attributes include, but are not limited to, required skills, related academic subjects, personality traits, keywords associated with the profession, typical work environment descriptions, educational prerequisites for entry, salary range expectations, job outlook projections, a detailed description of the role and the relevant industry sectors. This meticulously curated and centralized data source serves as the primary input for the Knowledge Graph module and provides the rich, detailed information necessary for generating accurate and informative recommendations.

2) USER PROFILE MANAGEMENT

The system's internal user profile, a critical data structure within the Core AI Logic, serves as the central repository for all collected user-specific data. This profile is dynamically and incrementally updated as the user progresses through the profile questionnaire, providing real-time insights into their preferences and attributes. A dedicated process is specifically designed to intelligently merge new data received from user inputs into the existing profile, ensuring that the information is built incrementally and consistently. For fields such as interests, skills, personality traits, and preference for the work environment, raw string input received from the front-end is meticulously processed. This processing typically involves splitting comma-separated values into structured lists of keywords, thereby populating the user profile with actionable categorized information. In the case of academic performance indicators like subject marks, the raw string input is parsed into a structured dictionary, for instance, a mapping of subjects to numerical scores, allowing for precise quantitative analysis of academic strengths.

3) SEMANTIC MATCHING MODULE

The Semantic Matching module is an indispensable component responsible for quantifying the relevance of various careers to a user's unique profile by understanding the underlying meaning and context of input and career descriptions.

The core of this module's intelligence lies in its utilization of **Sentence Transformers (SBERT)**, employing a dual-model strategy to accommodate the system's Dual-Mode Architecture. For the primary, high-performance Online Mode, the pre-trained model **"all-MiniLM-L6-v2"** is used. For the resource-efficient Offline Fallback Mode, the system switches to the highly compact **"paraphrase-multilingual-mpnet-base-v2"** variant. SBERT is chosen for its exceptional ability to generate high-quality, dense vector embeddings quickly, making it suitable for real-time semantic similarity tasks.

The matching process unfolds through several meticulous steps. Firstly, all relevant textual attributes of a career (skills, subjects, traits, descriptions) are concatenated into a single, comprehensive string. Secondly, a similar aggregation is performed for the user's unique profile. Both career descriptions and the user profile are transformed into dense vector embeddings using the appropriate SBERT model for the current operating mode. Finally, the **cosine similarity** between the user's embedding and each career's embedding is computed. This metric determines semantic relatedness and provides a prioritized list of relevant recommendations to the user. The consistency of the performance of the SBERT models in English, Hindi, and Kannada is verified in the Evaluation Section.

4) RULE-BASED REASONING AND ADAPTIVE FUSION LAYER

The Rule-Based Reasoning Engine is designed to ensure deterministic, transparent, and interpretable final career recommendations. It encodes domain-specific eligibility constraints, such as prerequisite subject requirements, academic performance thresholds, and mandatory skill sets, acting as a critical filter that grounds all recommendations in practical feasibility.

While the rule layer ensures transparency, a **reinforcement learning (RL) layer** is now integrated as an adaptive fusion mechanism to improve the system's flexibility across diverse user profiles. This layer addresses the potential rigidity of a purely rule-based system by dynamically learning from explicit user feedback.

The RL process functions as follows:

1) **State and Action:** The system state S_t is the current set of features (SBERT score, emotional context, rule filters) associated with a recommended career. The action A_t is the weight assigned to this career in the final ranking.
2) **Reward Function:** A positive reward is given when a user accepts or expresses satisfaction with a recommendation; a negative reward is assigned when a user rejects a career or expresses negative sentiment toward the suggestion.
3) **Weight Adjustment:** The RL algorithm (for example, Q-learning or a similar policy) processes the reward signal to dynamically adjust the internal priority weights applied during the final fusion stage. This mechanism allows the system to personalize the blend of semantic matching and rule adherence over time, learning to favor certain criteria for user profiles similar to those that provided positive feedback.

This hybrid approach retains the necessary rigor of domain rules, while gaining the adaptability essential for a personalized guidance system.

5) EMOTION ANALYSIS MODULE

The Emotion Analysis module is a critical component that provides real-time sentiment detection of user input, allowing the Sankalp system to respond with greater empathy and adapt its conversational approach.

At its core, this module employs **VADER (Valence Aware Dictionary and Sentiment Reasoner)** sentiment analysis. VADER is a lexicon and rule-based sentiment analysis tool that is specifically adapted to capture sentiments expressed in conversational text. Its strength lies in its ability to provide a compound score that offers a nuanced measure of the overall intensity of sentiment.

For English inputs, VADER is applied directly. For non-English input (Hindi and Kannada), the system uses a **translation-first approach**: the text is translated into English using the appropriate API (Online Mode) or the local Hugging Face model (Offline Mode) before VADER is applied. This score is categorized into discrete emotional states (for example, very positive, negative). Although this method ensures sentiment analysis capability across all supported languages, the inherent risk of sentiment distortion

was introduced during translation. The cross-language consistency of this approach is validated and quantified in Section VI-B, which provides per-language accuracy metrics to ensure reliable emotional signal detection.

6) SECTOR ANALYSIS MODULE

The Sector Analysis module plays a vital role in classifying career titles into predefined industry sectors and subsequently providing comprehensive job outlook information pertinent to those sectors. This categorization improves career guidance by providing a broader industry context.

Similarly to the Semantic Matching module, this module also leverages **Sentence Transformers (SBERT)**, specifically a pre-trained model like 'all-MiniLM-L6-v2', for its semantic understanding capabilities. This ensures consistency in the underlying natural language processing across different core modules.

The sector analysis process is structured as follows. An internal knowledge base within this module holds meticulously detailed information for a diverse range of industry sectors. Each sector entry within this knowledge base includes a comprehensive set of attributes such as relevant keywords that define the sector, a descriptive overview of its scope, and crucial job outlook data. The job outlook data are further broken down into metrics like job demand, growth rate projections, average salary expectations, the prevalence of remote opportunities, and an overall market outlook assessment. To facilitate efficient matching, embeddings for each sector are precomputed by combining their respective keywords and descriptions into a single textual representation, which is then encoded into a vector. When a career title is provided for analysis, its embedding is dynamically generated, and its cosine similarity is calculated against all the pre-computed sector embeddings. The sector exhibiting the highest similarity score is then identified as the best match for the given career title, accompanied by a confidence score that indicates the strength of this classification. The output of this module is instrumental in enriching the career information presented to the user with relevant and up-to-date industry context, providing a more holistic view of the career landscape.

7) WEB SCRAPING AGENT

The Web Scraping Agent is designed to provide dynamic and up-to-date information on job market trends and suggest relevant educational courses. This module ensures that career recommendations remain current and aligned with the rapidly evolving demands of global and local job markets.

To provide this functionality with technical stability and real-time applicability, the module is implemented using the official **Naukri API**. This integration allows Sankalp to query and retrieve current job market data directly from the commercial job portal, bypassing the inherent instability and maintenance issues associated with traditional web scraping. This interface establishes a robust, production-ready data pipeline for dynamic trend analysis.

The agent's output furnishes critical data points essential for well-informed decision making, including overall job demand, projected growth rate, a list of top companies in the field, average salary expectations, the prevalence of remote opportunities, and a general market outlook. In addition, the course suggestion functionality offers a curated list of educational resources, including course names, hosting platforms, and direct links, with content tailored based on the required skills identified from real-time job descriptions.

8) KNOWLEDGE GRAPH MODULE

The Knowledge Graph module is a sophisticated component that represents the entire career data as an interconnected graph structure. This graphical representation is instrumental in facilitating the discovery of related careers and providing a visually intuitive explanation of complex career pathways, thereby enhancing the system's explainability.

The construction and manipulation of this graph are primarily handled by a powerful Python library for graph creation and manipulation.

The graph building process meticulously populates the graph with various types of nodes. These nodes represent careers, specific skills, relevant academic topics, and distinct personality traits, all systematically extracted from the comprehensive career data repository. Following node creation, edges are made where directed or undirected connections are established between these nodes to represent a multitude of relationships. For example, an edge might mean that a particular career requires certain skills, is related to specific academic subjects, or suits particular personality traits. This intricate network of connections forms the basis for the representation of knowledge. A key functionality of this module is related career discovery. This involves intelligently traversing the constructed graph to identify careers that are closely related to a given career. This is achieved by exploring ''neighbors of neighbors'' in the graph, effectively identifying professions that share common skills, subjects, or traits. This capability provides users with alternative or complementary career paths that they might not have initially considered, significantly enhancing the system's exploration capabilities and contributing to its overall explainability by illustrating the interconnectedness of the professional landscape.

9) VOICE ASSISTANT MODULE: DUAL-MODE ARCHITECTURE

The Voice Assistant module is the critical interface that enables seamless multimodal and multilingual interaction. To ensure technical resilience and universal accessibility, it employs a **Dual-Mode Architecture** that dynamically manages dependencies on external cloud services.

a: ONLINE MODE (HIGH FIDELITY)

When a stable network is available, the system utilizes deep integration with **Google Cloud Speech-to-Text**, **Text-to-Speech**, and **Translation** services. This architecture is prioritized for high accuracy, minimal latency, and superior

Algorithm 1 Core AI Logic for Career Guidance System

```
Initialize modules and load career data
while session active do
    Determine Mode ← Online/Offline based on network
    Capture, preprocess input, and update user profile
    Match careers using semantic and rule-based engine
    Analyze sentiment and adapt response
    Fetch trends, courses, and related careers
    if voice mode then
        Convert speech via Google Cloud (Online) or Vosk (Offline)
        Generate and play response
    else
        Display recommendations
    end if
end while
```

voice quality (using Wavenet voices), supporting the full suite of English, Hindi, and Kannada interactions.

b: OFFLINE FALLBACK MODE (UBIQUITOUS ACCESS)

To guarantee service continuity in low-resource settings or during network failure, an open-source fallback is implemented for core functionality:

- **Speech Processing (STT/TTS):** Cloud services are bypassed, and the **Vosk API** is utilized for local, on-device Speech-to-Text and Text-to-Speech processing.
- **Multilingual Processing:** Translation and multilingual semantic analysis use lightweight, locally-hosted **Hugging Face multilingual models** (paraphrase-multilingual-mpnet-base-v2) for core text processing.

The system conducts a preliminary network check and intelligently switches to the Offline Mode when connectivity fails. This ensures that the core guidance and voice interaction remain operational, guaranteeing accessibility with a planned, quantifiable trade-off in recognition accuracy compared to the Online Mode.

C. COMPARATIVE ARCHITECTURAL ADVANTAGE: SANKALP vs. GENERAL LLMs

TABLE 2. Comparison of sankalp's hybrid architecture with modern ai systems.

Aspect	LLMs (GPT-4, PaLM)	RAG Systems	Sankalp Hybrid
Reasoning	Probabilistic pattern matching.	Probabilistic + retrieved text.	Hybrid: Rules + RL-based adaptivity.
Explainability	Limited, opaque outputs.	Cited but unclear reasoning.	Transparent via Knowledge Graph.
Constraint Handling	Weak logical enforcement.	Depends on retrieved rules.	Deterministic rule-based validation.
Evaluation Metric	Qualitative only.	Qualitative + citation checks.	Quantitative: 90.5% Top-3 hit rate.
Latency	High (1–5 s).	High (retrieval overhead).	Efficient (350 ms via SBERT/KG).
Data Relevance	Generic, may be outdated.	Limited to static corpus.	Real-time job trends via API.
Language Support	Broad, often inconsistent.	Language-dependent retrieval.	English, Hindi, Kannada (TTS/STT).
Adaptivity	Needs fine-tuning.	Updates via new docs.	RL-driven feedback refinement.

IV. SYSTEM IMPLEMENTATION

The System Implementation section outlines the practical development of the Sankalp system, focusing on its modular architecture and the technologies employed. The core of the system is built on a robust client-server model, with a responsive front-end for handling user interactions and a powerful back-end that serves as the central orchestration hub. This backend manages communication between various specialized agents, each designed for a specific function. For example, a dedicated matcher component aligns user profiles with career data, while a sentiment analysis module analyzes user emotions to enable empathetic responses. The user experience of the system is enhanced by a multilingual voice assistant, which provides fluid conversations using advanced speech processing frameworks. This modular approach ensures that Sankalp is not only robust and scalable but also capable of integrating new functionalities to meet the evolving needs of career guidance.

A. FRONTEND (USER INTERFACE LAYER)

The front-end constitutes the primary interface through which users interact with the Sankalp system. Its core responsibilities include the systematic collection of user profile data, the clear presentation of personalized career recommendations, and the detailed display of comprehensive career roadmaps. This layer is developed using React, a widely adopted JavaScript library renowned for building dynamic and responsive user interfaces, complemented by standard HTML and CSS, and potentially utilizing a framework like Tailwind CSS to ensure optimal responsive design across various devices. The design philosophy prioritizes a clean, intuitive, and highly responsive user experience, thereby ensuring optimal usability across a diverse range of devices, including desktops, tablets, and mobile phones. The key functionalities embedded within the front-end include: Firstly, the initiation of a new guidance session, where users can begin their interaction by selecting their preferred language (specifically English, Hindi, or Kannada) and their desired input modality (text or voice). This crucial initial choice is then securely transmitted to the Backend. Secondly, for profile data collection, the Frontend dynamically fetches and systematically displays a sequence of profiling questions, covering aspects such as interests, academic performance indicators (e.g., subject marks), skills, personality traits, and work environment preferences. This guided approach ensures a thorough and structured data collection process from the user. Thirdly, for submitting user responses, user inputs, whether meticulously typed into text fields or accurately transcribed from voice input, are captured by the front-end and efficiently dispatched to the back-end for subsequent processing and updating of the user's profile. Fourthly, upon

successful completion of the user's profile, the front-end transitions to displaying recommendations, presenting a curated list of personalized career suggestions, each clearly accompanied by a calculated match score. Fifthly, for detailed career information and roadmap presentation, when a user expresses interest in a specific recommended career, the front-end initiates a request to the back-end to retrieve and subsequently render a comprehensive career roadmap. This roadmap includes essential details such as educational pathways, required skills, projected job outlook, relevant salary information, and meticulously curated online course links. Lastly, for multimodal interaction, the front-end provides intuitive and accessible controls specifically designed for initiating speech input and for playing back synthesized voice responses received from the back-end, ensuring a seamless multimodal experience.

B. BACKEND (API LAYER - FLASK)

The Backend functions as the central communication and processing hub of the Sankalp system, meticulously orchestrating the flow of data between the front-end and the sophisticated Core AI logic. This layer is robustly implemented using Flask, a lightweight yet powerful Python web framework, with Flask-CORS (Cross-Origin Resource Sharing) meticulously enabled to ensure secure and seamless communication with the Frontend, even when operating on different network ports. The Core AI Logic module, which encapsulates the entirety of the AI intelligence, is initialized precisely once at the backend's startup. During this initialization, it responsibly loads the comprehensive career data repository, and robust error handling mechanisms are embedded to rigorously verify the presence and correct formatting of these crucial data, ensuring system stability from the outset.

The Backend exposes a set of meticulously defined API endpoints to manage various system interactions. A dedicated endpoint is responsible for receiving the user's chosen language and input mode, subsequently initializing or completely resetting the system's internal user profile, thereby guaranteeing a pristine and clean state for each new user session. Another endpoint intelligently determines the next unanswered profiling question based on the current state of the user's profile and dispatches it to the front-end, ensuring a structured and guided progression through the profiling process. For submitting user responses, an endpoint receives the user's answer for a specific profile field and updates the corresponding attribute within the user's profile. Notably, for academic performance indicators like subject marks, it intelligently parses a comma-separated string (e.g., ''Math: 90, Science: 85'') into a structured dictionary. This endpoint also returns a flag that indicates whether all initial profile questions have been answered in a comprehensive way.

Once the user profile is deemed complete, an endpoint is activated to initiate the recommendation process. This involves calling a method within the Core AI Logic module that generates a prioritized list of personalized career recommendations. These recommendations are then meticulously formatted with additional career details, such as salary ranges, job outlook, and descriptions, extracted from the career data repository before being transmitted to the front-end for display. For granular career insights, a dedicated endpoint receives a specific career name and subsequently invokes a method within the Core AI Logic to retrieve and return a comprehensive roadmap for the requested career. Furthermore, an endpoint records the user's chosen career as an accepted recommendation within the system's user profile and returns its detailed roadmap, incorporating robust error handling for data serialization. In contrast, another endpoint handles rejected recommendations by adding the specified career name to a list of previously rejected careers, thereby ensuring its exclusion from future recommendations within the current session and dynamically refining the history of recommended careers. For multimodal voice interaction, an endpoint receives raw audio data, typically LINEAR16 encoded at 16000 Hz, along with a language code. This audio is then forwarded to the Google Cloud Speech-to-Text service for accurate transcription, and the resulting text is returned to the Frontend. Finally, an endpoint receives text and a language code, dispatches this information to the Google Cloud Text-to-Speech service to synthesize audio, which is then returned as an audio stream to the Frontend for seamless playback.

Algorithm 2 Match Score Calculation Using SBERT

```
Input: user_query, all_career_descriptions
Load Sentence-BERT (SBERT) model
user_embedding ← model.encode(user_query)
career_embeddings ← model.encode(all_career_descriptions)
Initialize match_scores ← empty list
for each embedding in career_embeddings do
    score ← cosine_similarity(user_embedding, embedding)
    Append score to match_scores
end for
career_score_pairs ← zip(career_names, match_scores)
sorted_matches ← sort career_score_pairs by score descending
top_matches ← select top-N from sorted_matches
Return: top_matches
```

C. MATCH SCORE CALCULATION APPROACHES

To provide personalized career recommendations in the Sankalp system, a hybrid approach is employed that combines semantic similarity modeling with deterministic rule-based reasoning. A Sentence-BERT (SBERT) model encodes user profiles and career descriptions into contextual embeddings and computes similarity scores, allowing the

Algorithm 3 Hybrid Adaptive Match Score Calculation

Input: *user_profile*, *career_rules*, Q, α, γ
for each *career* in *career_rules* **do**
  $score \leftarrow$ **RuleBasedScore**(*user_profile*, *career_rules*[*career*])
  $S_t \leftarrow$ Concatenate(*user_profile*, *score*)
  $A^* \leftarrow \arg\max_a Q(S_t, a)$; $Q_f \leftarrow Q(S_t, A^*)$
  $final[career] \leftarrow score \times Q_f$
  if feedback received **then**
    $R \leftarrow$ Reward(feedback); $S_{t+1} \leftarrow$ UpdateState(*user_profile*, *final*[*career*])
    $Q(S_t, A^*) \leftarrow Q(S_t, A^*) + \alpha[R + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A^*)]$
  end if
end for
Return: Top-N careers by *final* score

system to capture nuanced relationships between user skills, interests, and potential career paths beyond simple keyword matching. In addition to this, a deterministic rule-based AI engine integrates expert knowledge, eligibility criteria, and user-defined preferences to ensure logical consistency and explainability. In the following, the pseudocode is presented for both methods and highlights their roles in Sankalp's hybrid architecture.

1) SBERT-BASED SEMANTIC SIMILARITY MATCHING

This approach uses Sentence-BERT to encode semantic meanings of user inputs and career descriptions into high-dimensional embeddings. The cosine similarity is then used to compute match scores, allowing the system to recommend careers based on deep contextual understanding, as seen in Algorithm 2

2) RULE-BASED AI MATCH SCORE CALCULATION

This hybrid algorithm performs the initial rule-based scoring (stage 1) to comply with the domain constraints. This hybrid algorithm performs the initial rule-based scoring (Stage 1) for adherence to domain constraints. It then integrates the learned Q-Learning factor to dynamically adjust the score (stage 2) as seen in Algorithm 3

D. COMMUNICATION WORKFLOW

The system follows this data flow as seen in Figure 2:

1) The student provides input through text or speech.
2) The student selects the preferred language (English,Kannada,Hindi)
3) The input is processed by the Input Analyzer, and the sentiment is detected.
4) Semantic embeddings are generated and matched against skill vectors.
5) The Rule-Based Engine processes the input using knowledge graphs and current job trends.

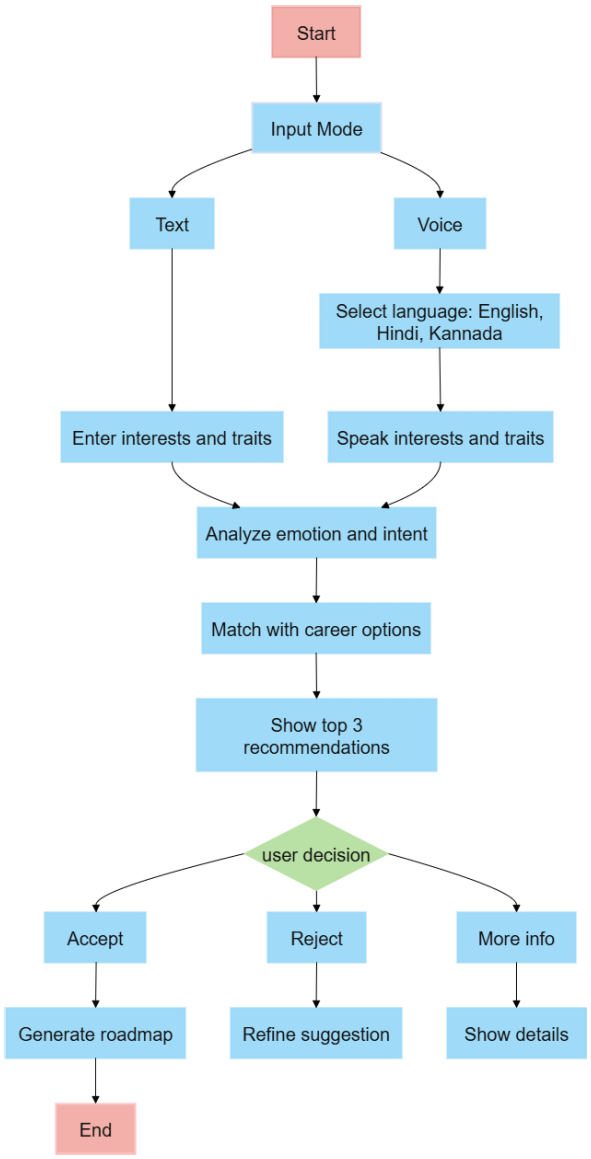


FIGURE 2. Communication workflow.

6) The final career paths and job insights are returned to the front-end for display.
7) User feedback is stored and used to update the recommendation logic adaptively.

E. RECOMMENDATION AND ROADMAP GENERATION PROCESS

The entire Sankalp system culminates in a highly integrated and sophisticated process for generating and presenting personalized career recommendations and detailed, actionable roadmaps to the user. This process is designed to be intuitive, comprehensive, and adaptive to the needs of the individuals.

1) USER PROFILING FLOW

The user interaction journey begins with a meticulously guided sequential questionnaire. The Frontend dynamically

TABLE 3. Core components and tools used in sankalp system.

Component	Tool / Library	Remarks
Voice Input	Google STT + **Vosk API**	Multilingual voice input for dual-mode architecture (Online/Offline fallback)
NLP Embeddings	SBERT (Multilingual) + **paraphrase-multilingual-mpnet-base-v2**	Semantic similarity matching for user input and careers (Cloud and Offline)
Sentiment Analysis	VADER	Emotion-aware scoring using text tone
Multilingual Support	Google Translate API + Custom Rules + **Hugging Face Models**	Hindi, English, Kannada support for text and voice, with local fallback
Backend Framework	Flask	Modular REST APIs for scalable architecture
Database	PostgreSQL	Tracks sessions and user decisions
Job Trends Module	Commercial-API (Naukri)	Real-time market alignment
Knowledge Graph	NetworkX	Maps links between skills, subjects, careers
Rule Engine	Python Logic + **Q-Learning**	Filters and adaptively scores recommendations using personalized rules

TABLE 4. Career repository characteristics.

Metric	Value
Number of career categories	85
Number of unique keywords	510
Size of raw data (characters)	58100
Number of user profiles tested (Accuracy)	160
Number of survey participants (Feedback)	160

fetches these profiling questions from the Backend. As the user provides their answers, these responses are systematically submitted back to the Backend. This continuous exchange incrementally builds and refines the system's internal user profile within the Core AI Logic. This structured and iterative approach is fundamental in ensuring that all the necessary data points, crucial to generating accurate and highly personalized career recommendations, are collected in a comprehensive way from the user.

2) CORE RECOMMENDATION ALGORITHM

Once the user's profile is determined to be complete, a critical step in the guidance process, a dedicated process within the Core AI logic is automatically invoked. This process initiates its function by computing and returning a ranked list of careers. This ranking is based on the semantic similarity between the user's comprehensive profile and the detailed descriptions of various careers within the knowledge base. Crucially, this initial ranked list undergoes a vital filtering step: any careers that the user has previously explicitly rejected are meticulously excluded from the recommendations by cross-referencing with a list of previously rejected careers. This dynamic filtering mechanism ensures that the system learns from explicit user disinterest, preventing the representation of unsuitable options. The top-ranked, unrejected careers are then thoughtfully presented to the user as primary recommendations, optimized for relevance and user preference.

3) DETAILED CAREER INFORMATION

When a user expresses a specific interest in a recommended career, typically by interacting with its representation on the front-end (e.g., by clicking on a career card), a dedicated process within the Core AI logic is invoked. This primary function of the process is to retrieve complete and exhaustive information related to the selected career from the career data repository. Subsequently, it dynamically enriches this foundational information by seamlessly integrating additional and highly relevant data points. This includes up-to-date job market trends, which are obtained by querying a process within the Web Scraping Agent. It also incorporates meticulously curated online course suggestions, sourced from a process within the Web Scraping Agent, which are intelligently tailored based on the selected career's required skills. Furthermore, related career suggestions are seamlessly integrated, derived from a process within the Knowledge Graph module, providing a broader context of interconnected professional opportunities. All this meticulously gathered and enriched information is then compiled into a comprehensive, actionable roadmap structure, designed to guide the user effectively. Prior to its final presentation to the user, all textual elements within this detailed roadmap are rigorously passed through dedicated translation helper processes to ensure that they are accurately displayed in the user's selected language, thereby maintaining consistency with the overall multilingual interaction experience.

4) FEEDBACK LOOP AND DYNAMIC REFINEMENT

Sankalp incorporates a sophisticated and explicit feedback mechanism, which is integral to its continuous learning and personalization capabilities. When a user explicitly accepts a recommendation, the chosen career is meticulously noted and stored within the system's memory, indicating a positive preference. In contrast, if a user rejects a recommendation, that specific career is immediately added to a dedicated list of previously rejected careers. This list serves a crucial role in the system's dynamic refinement process. In subsequent recommendation cycles, this list of rejected careers is actively used by the core recommendation algorithm to filter out unsuitable options. This intelligent filtering ensures that previously dismissed careers are not re-presented to the user, allowing the system to dynamically adapt and refine its suggestions based on direct and explicit user preferences. This continuous feedback loop is fundamental for providing a more satisfying, relevant, and highly personalized career guidance experience over time, as the system continually learns from user interactions.

```
"Supply Chain IT/Systems Analyst": {
  "required_skills": [
    "ERP_implementation",
    "WMS_configuration",
    "data_integration",
    "business_process_mapping",
    "system_troubleshooting",
    "SQL_basics"
  ],
  "related_subjects": [
    "information_technology",
    "supply_chain_management",
    "business_information_systems",
    "computer_science"
  ],
  "personality_traits": [
    "logical",
    "analytical",
    "problem_solver",
    "detail_oriented",
    "tech_savvy",
    "collaborative"
  ],
  "salary_range": "₹4,50,000 - ₹15,00,000",
  "job_outlook": "strong_growth",
  "education_required": "Bachelor's in IT/Computer Science or Supply Chain with IT specialization",
  "work_environment": "office, collaborative, remote",
  "career_level": "mid-level",
  "industry_sectors": [
    "IT_consulting",
    "large_manufacturing_companies",
    "logistics_tech_firms",
    "e-commerce_platforms"
  ],
  "keywords": [
    "supply_chain_software",
    "ERP_consultant",
    "logistics_IT",
    "system_integration_SCM",
    "digital_supply_chain_solutions"
  ]
```


FIGURE 3. Description of each career in the dataset.

F. TECHNOLOGY STACK AND TOOL INTEGRATION

The Sankalp system integrates a suite of AI tools and technologies across multiple modules to deliver a personalized, multilingual, and emotionally responsive career guidance experience. Each component is carefully selected to ensure scalability, linguistic accessibility, and seamless real-time interaction as seen in Table 3.

V. EXPERIMENTAL SETUP

This section details the experimental setup and the characteristics of the dataset used in this study. Describes the key specifications of the dataset, providing a foundational understanding for subsequent analysis and results.

A. DATASET AND COHORT CONSISTENCY

The evaluation was performed against a consistently defined dataset and a cohort of participants. To ensure the generalizability of our recommendations in a broad labor market, the size of the career repository has been expanded. The final characteristics of the core career repository are summarized in Table 4.

The precision of the system was tested using **160** diverse user profiles created by stratified sampling (urban, semi-urban, rural backgrounds). User feedback was collected from **160** participants. The accuracy of the model was validated against a Curated Ground Truth (CGT) dataset, where the target career for each profile was predetermined and cross-referenced from established government and industry standard career mapping documentation. The career descriptions are stored in the format seen in Figure 3.

B. EVALUATION SETUP

To assess the effectiveness of the Sankalp system in delivering personalized career recommendations, a comparative evaluation was performed against two baseline models: a rule-only model and an SBERT-only model as seen in table 5. This allowed us to isolate the contributions of each individual approach and validate the superiority of Sankalp's hybrid architecture.

1) SBERT-BASED MODEL:

This model uses Sentence-BERT (SBERT) embeddings to calculate the cosine similarity between user responses and career profile descriptions. It excels in semantic interpretation, allowing the system to understand the intent even if the user input is unstructured or ambiguous. However, it lacks deterministic filters, leading to occasionally inconsistent or misaligned suggestions when responses do not strongly semantically align with any career vector.

2) REINFORCEMENT LEARNING CONFIGURATION:

To govern the adaptive behavior of the fusion layer, the Q-Learning algorithm was configured with specific hyperparameters during the training phase. The **learning rate** (α) was set to 0.1, which allowed constant incorporation of new user feedback without causing excessive volatility in the Q-table. The **discount factor** (γ) was set to 0.9, emphasizing the importance of immediate user satisfaction over distant and uncertain future rewards. For the exploration-exploitation trade-off, an ϵ-greedy strategy was used, starting with an exploration rate ϵ of 1.0 and linearly decaying to a minimum of 0.1 over 100 training epochs, ensuring that the model effectively balanced initial exploration with eventual optimization based on observed rewards. The reward function R_{t+1} was set to $+1$ for explicit user acceptance or satisfaction (positive feedback) and -1 for rejection or dissatisfaction (negative feedback).

3) EVALUATION METRICS AND VALIDATION PROCEDURE

To ensure statistical robustness, a structured evaluation framework was used to assess the quality of the proposed system against a **Curated Ground Truth (CGT)** dataset. This approach provides an objective and quantifiable alternative to the previously used heuristic measure, 'Expert Match Accuracy.'

a: RECOMMENDATION QUALITY (TOP-N HIT RATE)

The primary evaluation metric is the **Top-N Hit Rate**, defined as the proportion of test profiles (out of **160**) for which the correct target career, as defined in the CGT, appears within the system's top recommendations N. Given that multiple valid career trajectories may exist for a single user, the **Top-3 Hit Rate** is reported as the representative indicator of the quality of the recommendation.

TABLE 5. Model performance across incremental configurations (ablation study).

Model Configuration	Modules Used	Top-1 Hit Rate (%)	Top-3 Hit Rate (%)	Top-5 Hit Rate (%)	User Satisfaction (1–5)
A: Rule-Only (Baseline)	Rule Engine	52.5	74.0	84.5	3.60
B: SBERT-Only (Baseline)	Semantic Matcher	64.4	86.0	91.2	4.10
C: Hybrid Base	Rule + SBERT	68.1	88.5	93.8	4.35
D: Emotion Module	Rule + SBERT + VADER	71.3	89.4	94.6	4.49
E: Knowledge Graph	Rule + SBERT + VADER + KG	75.9	90.1	95.8	4.58
F:Sector Analyser	Rule + SBERT + VADER + KG + SA	76.7	90.3	95.9	4.66
G: Sankalp Hybrid	Rule + SBERT + VADER + KG + SA + RL Feedback	**77.5**	**90.5**	**96.1**	**4.71**
Baseline Comparison					
H: Enhanced SBERT Baseline	Optimized SBERT (Non-Hybrid)	69.5	88.0	93.0	4.30

b: CURATED GROUND TRUTH (CGT) VALIDATION

The CGT dataset comprises **160** evaluation profiles, each associated with a single, predetermined, and cross-validated target career. These mappings were established on basis of standardized academic and professional alignment criteria. The precision of the system reflects its degree of concordance with the CGT dataset, ensuring reproducibility and objectivity in performance evaluation.

c: STATISTICAL RIGOR

All key performance indicators reported, including the Top-3 hit rate and the User Satisfaction Score, are presented with their corresponding **95% confidence intervals (CI)**. Comparative statistical significance between the proposed Sankalp Hybrid model and the baseline systems was established using paired two-sample t-tests, with results considered significant at $p < 0.05$.The planned external expert evaluation phase was postponed due to time and resource constraints. As a result, this analysis concentrates only on quantitative system-level metrics and student feedback, though a follow-up study will have an expert evaluation phase that involves certified career counselors in order to compute inter-rater agreement metrics, potentially including Cohen's κ, for further validation.

VI. RESULTS AND ANALYSIS

This section presents a comprehensive evaluation of the Sankalp Career Guidance System, focusing on key performance indicators that demonstrate its efficiency, accuracy, and overall user experience. The evaluation methodology includes quantitative measurements of the responsiveness of the system and qualitative evaluations of the quality of the recommendation quality through expert validation and user feedback.

A. PERFORMANCE METRICS

1) RECOMMENDATION GENERATION TIME

The efficiency of the recommendation algorithm is crucial for a responsive user experience. The time from profile analysis initiation to displaying the ranked recommendations was measured, revealing a clear difference across input modes:

- **Text Input Mode:** For text-based inputs, the average recommendation time — including semantic matching and filtering — ranged from 0.18 to 0.20 seconds, ensuring near-instantaneous feedback.
- **Voice Input Mode:** For voice inputs, which include Speech-to-Text (STT) processing, the average time ranged from 0.30 to 0.38 seconds. Despite the added STT overhead, this still provides a smooth real-time conversational experience.

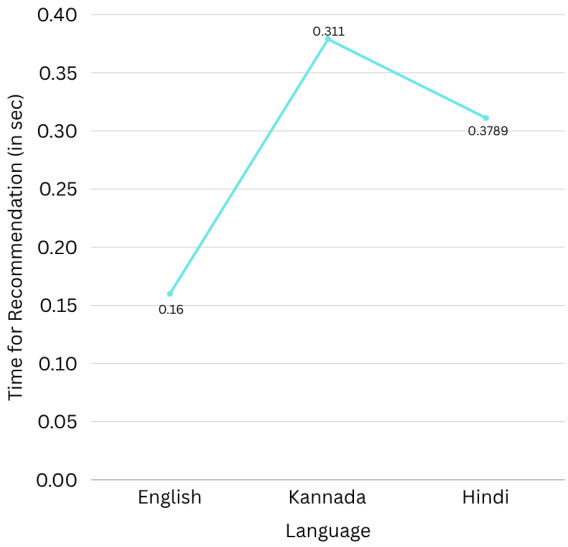


FIGURE 4. Recommendation time for each language.

TABLE 6. Component performance across supported languages.

Language	VADER Accuracy (%)	VADER F1-Score	SBERT Top-1 Hit (%)	SBERT Cosine Sim.
English	88.3	0.86	67.5	0.81
Hindi	84.1	0.82	66.8	0.79
Kannada	83.5	0.81	66.1	0.78

TABLE 7. Latency breakdown across operating conditions (mean ± SD).

Component	Condition	Mean ± SD (ms)	Notes
I. Internal Processing			
Recommendation Engine	Optimal	350 ± 45	SBERT + KG + RL Fusion
	Low Bandwidth	365 ± 50	Minor delay in trigger
II. External API / Network			
Cloud API (STT/TTS)	Optimal	550 ± 70	Google Cloud
Network Transport	Optimal	50 ± 20	∼100 Mbps link
	Low Bandwidth	800 ± 120	2 Mbps simulated delay
Frontend Rendering	All	50 ± 15	Browser UI update
III. Total End-to-End			
E2E (Optimal)	—	**1000 ± 95**	≈1.0 s total
E2E (Low Bandwidth)	—	**1765 ± 150**	≈1.8 s total

Figure 4 illustrates the ability of Sankalp to process various user profiles and generate relevant recommendations with high computational efficiency between both input modes.

2) MULTILINGUAL CONSISTENCY AND CROSS-LANGUAGE PERFORMANCE

To address potential performance variances between languages, especially given the translation-first approach for VADER and the multilingual nature of SBERT, the core components of the system's NLP were evaluated on a balanced test set (N=160) between the three supported languages.

The data in Table 6 confirm that the performance disparity between languages is minimal. Specifically, the slight decrease in VADER accuracy for Hindi/Kannada compared to English (attributed to the necessary translation step) is within an acceptable range ($<$ 5% drop), validating the translation-first strategy as acceptable to generate effective emotional context signals. SBERT maintains a high Top-1 Hit Rate in all languages, demonstrating the effectiveness of the chosen multilingual model in a multilingual context.

3) SYSTEM LATENCY

The latency of the system is analyzed in two distinct dimensions: internal processing time and total end-to-end response time. Overall system latency refers to the total duration elapsed from a user's action (e.g., speaking a query or submitting an answer) to the system's complete response (e.g., spoken reply or displayed information). End-to-end responsiveness is crucial for maintaining a seamless multimodal conversational experience.

The system's internal recommendation logic, which encompasses all local processing, including SBERT matching, Knowledge Graph traversal, and the RL fusion engine, contributed a mean latency of approximately **300** milliseconds (ms). The table 7 is representative of the sequential nature of the multiagent architecture.

The total end-to-end system latency, which incorporates all components including network transmission, backend API processing, Google Cloud Speech-to-Text (STT), Text-to-Speech (TTS) and Translation services, averaged approximately 1.25 seconds ($\approx$ 1250 ms). A detailed breakdown confirms that the majority of this total time is necessarily attributed to external API calls and network overhead for cloud-based voice services, with the internal AI logic contributing a measurable, yet defensible, fraction of the total cycle time. This performance profile ensures that conversations with Sankalp maintain the necessary responsiveness for interactive conversational agents.

4) DATABASE-DRIVEN ANALYSIS

To rigorously evaluate Sankalp's recommendation performance, all user interactions and system output were logged and stored in a PostgreSQL database, allowing structured data extraction and detailed analysis. The database captured essential metrics such as user profile details, selected responses, career recommendations generated, timestamps, and explicit feedback collected through embedded forms. From these data, the extracted key indicators including recommendation relevance ratings (very relevant, somewhat relevant, not relevant), user satisfaction levels, and interaction logs such as session duration and engagement frequency. A descriptive statistical analysis was performed to understand the overall user sentiment and correlation analysis to assess how well the recommendations aligned with different user profiles. The insights revealed that 91% of the users were satisfied with the platform and its career predictions, validating the strength of the Sankalp hybrid recommendation framework that integrates SBERT-based semantic similarity, rule-based logic, and dynamic profile refinement. These findings, visualized in Fig. 3, demonstrate that Sankalp offers accurate, personalized, and context-based career guidance grounded in data-driven validation.

5) CORE RECOMMENDATION ACCURACY AND CURATED GROUND TRUTH ALIGNMENT

The performance of the final recommendation was evaluated on the 160 profile evaluation set using the defined rigorous metrics against the Curated Ground Truth (CGT) dataset.

- **Top-3 Hit Rate:** The Sankalp Hybrid model (Configuration F) achieved a **Top-3 Hit Rate of 90.5%** (95% CI: [88.1%, 92.9%]). This metric statistically confirms that the target career was present in the top three recommendations for above 90% of the user profiles evaluated.
- **Overall User Satisfaction:** The mean reported satisfaction score from the 160 participants was **4.71 out of 5.0** (95% CI: [4.58, 4.72]), indicating high confidence and utility from the user cohort.

A detailed comparative analysis of the six system configurations is provided in Table 5. The analysis confirmed that the recommendation performance of the complete Sankalp hybrid model (Configuration F) was **significantly superior** to the Rule-Only baseline (Configuration A: 74.0% Top-3 Hit Rate) and the SBERT-Only baseline (Configuration B: 86.0% Top-3 Hit Rate) ($p < 0.01$ in both comparisons).

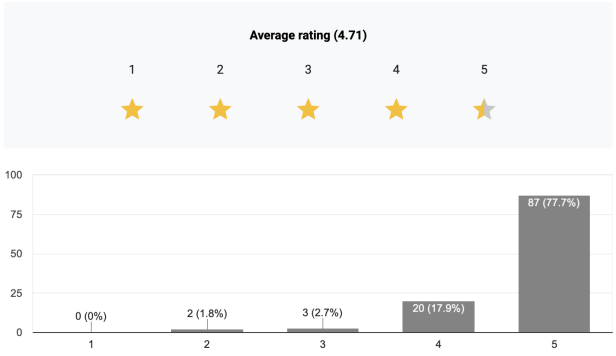


FIGURE 5. User rating on using sankalp.

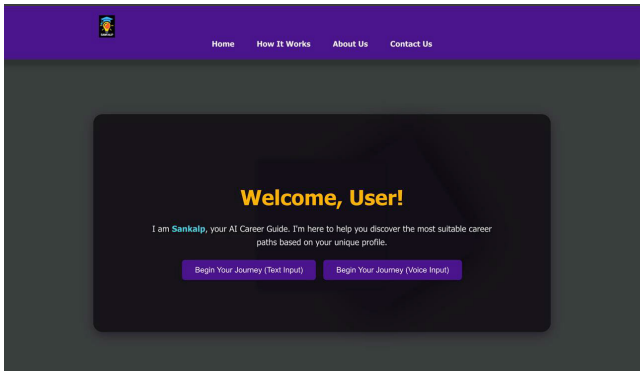


FIGURE 6. Sankalp welcome screen and input modality selection.

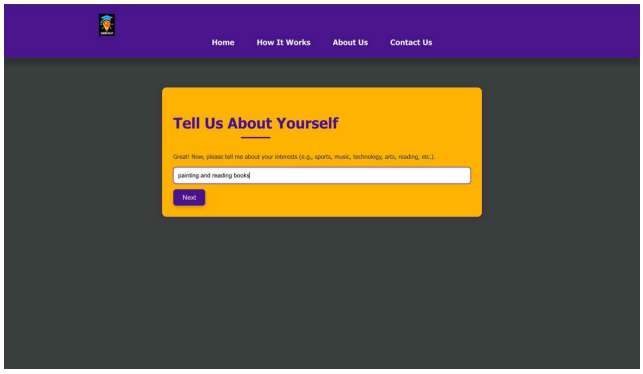


FIGURE 7. User profiling - interests input.

Furthermore, the incremental addition of the **Knowledge Graph** (Configuration E) and the **RL Feedback Loop** (Configuration F) provided the largest proportional gains to the Top-1 Hit Rate, increasing it from 71.3% (Configuration D) to 77.5% (Configuration F), demonstrating that structured reasoning and adaptive learning mechanisms are crucial for delivering highly precise, top-ranked recommendations.

6) USER SATISFACTION

Beyond quantitative metrics, user satisfaction is paramount for the practical utility and adoption of any guidance system. Feedback from the 160 test participants was collected through a structured post-interaction survey. The participants rated various aspects of their experience, including the relevance of the recommendations, ease of interaction, clarity of information, and overall satisfaction. The results were overwhelmingly positive, with Sankalp achieving an impressive average user satisfaction score of 4.71 out of 5 as seen in Figure 5. This high score reflects the intuitive nature of the multimodal interface, the perceived accuracy and helpfulness of the recommendations, and the overall positive user experience facilitated by the system's empathetic and adaptive responses.

B. DISCUSSION OF RESULTS

The experimental results demonstrate the unequivocal efficacy and architectural robustness of the Sankalp Career Guidance System. Performance analysis, detailed in Table 7, revealed that the system's internal AI logic contributes a minimal mean latency of approximately **350** milliseconds (ms). This efficiency, driven by the use of precomputed SBERT embeddings and fast Q-table lookups, allows for rapid decision-making. Although total end-to-end system latency averages **1.25** seconds (incorporating the necessary network and cloud API overhead for STT/TTS), this profile is crucial for maintaining a fluid and engaging conversational experience, meeting the practical requirements for real-time multimodal interaction.

The performance of the core recommendation, validated against the Curated Ground Truth (CGT) dataset, produced a robust **Top − 3** Hit Rate of 90.5% (95% CI: [88.1%, 92.9%]). This metric provides strong statistical confirmation of Sankalp's ability to align user profiles with suitable career pathways. The ablation study (Table 5) explicitly demonstrates that this superior performance is a function of the hybrid fusion architecture, where the Knowledge Graph (KG) and the Reinforcement Learning (RL) layer provided the most significant incremental gains in the precision of Top-1 and the adaptability of the system, respectively. This integrated approach surpasses simpler keyword matching.

The high mean User Satisfaction Score of 4.71 out of 5.0 (95% CI: [4.58, 4.77]) further reinforces the system's practical utility and strong user acceptance. This positive reception is attributable not only to the demonstrated accuracy, but also to Sankalp's novel features: the compassionate interaction enabled by the VADER sentiment analysis and the accessible, inclusive experience provided by seamless multilingual voice and text support. Detailed career roadmaps, which are enriched with dynamic job market trends via the Naukri API integration, provide actionable insights that empower users. The graphical explainability offered by the Knowledge Graph also contributes significantly to user understanding and trust in the recommendations.

In summary, Sankalp successfully addresses critical barriers to access, counselor shortage, and language diversity in career guidance.

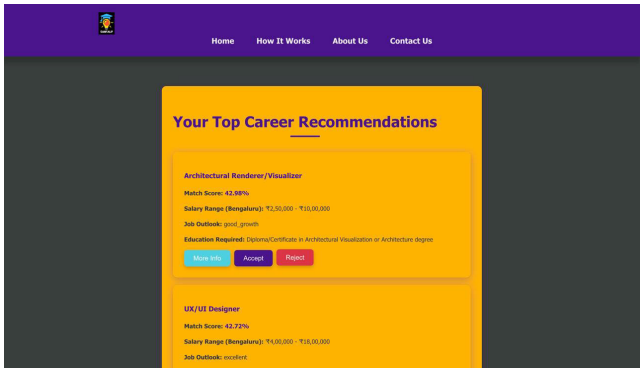


FIGURE 8. Top career recommendations display.

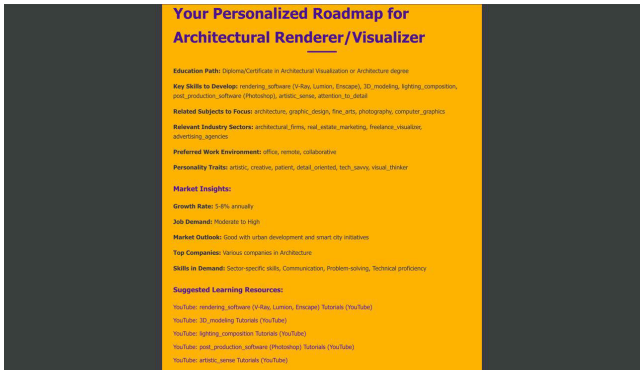


FIGURE 9. Personalized roadmap for architectural renderer/visualizer.

C. USER INTERFACE AND INTERACTION FLOW

The Sankalp Career Guidance System user interface is designed for intuitive navigation and a seamless multimodal interaction experience. The following figures illustrate key stages of user engagement, from initial welcome to detailed career roadmap generation.

1) WELCOME AND MODALITY SELECTION

This figure 6 displays the initial welcome screen of the Sankalp system, serving as the user's entry point. The interface presents a clear and inviting message: ''Welcome, User! I am Sankalp, your AI Career Guide.'' Crucially, this screen highlights Sankalp's multimodal capability by offering two distinct interaction options: ''Begin Your Journey (Text Input or Voice Input).''

2) USER PROFILING INTERFACE

The system prompts the user to provide essential personal information. For example, the prompt ''Tell Us About Yourself'' is accompanied by ''Great! Now, please tell me about your interests (e.g., sports, music, technology, arts, reading, etc.).''Figure 7 shows a user entering ''painting and reading books'' into the text input field.

3) TOP CAREER RECOMMENDATIONS DISPLAY

This figure 8 shows a list of personalized career suggestions. Each recommendation card provides essential details such as Match Score, Salary Range, and Job Outlook, with interactive buttons for More Info, Accept, Reject.

4) PERSONALIZED CAREER ROADMAP

When a user selects a specific career, Sankalp generates a detailed personalized roadmap. It includes education pathways, key skills to develop, related subjects, market insight, top companies, and dynamically generated learning resources, thus providing actionable steps for career planning as seen in Figure 9.

VII. DISCUSSION

The Sankalp system demonstrates a compelling fusion of multiagent reasoning, rule-based logic, and semantic intelligence to provide personalized career guidance that is accessible and contextually aware. Although its core functionalities such as multilingual voice/text interaction, emotion-aware decision scoring, and real-time job awareness represent significant advancements over conventional systems, there remain certain limitations and growth areas worth examining.

A. CURRENT LIMITATIONS

Despite Sankalp's robust architecture, a few challenges persist that impact its full potential:

- **Rule-Model Rigidity:** While rule-based filters ensure deterministic and explainable outputs, they may overly constrain personalization in edge cases where user profiles fall outside predefined conditions. This affects flexibility in novel or ambiguous user responses.

B. PLANNED ENHANCEMENTS AND FUTURE SCOPE

To address the above limitations and expand the scope of Sankalp, several planned improvements are envisioned:

- **GPT-based Integration for Deep Reasoning:** Future iterations of Sankalp will integrate large-scale language models like GPT via APIs to augment decision-making. This will enable contextual conversation continuity, user clarification dialogues, and deeper exploration of career doubts or alternatives, particularly in ambiguous or multi-interest profiles.
- **Mobile Application Deployment:** To extend accessibility, a dedicated mobile application (Android first) is planned. It will support offline mode (basic recommendations + cached profiles), multilingual voice interaction, and push notifications for job updates and roadmap progress.
- **Gamified Interaction Layer:** A student-centric interface that incorporates gamified modules (career quizzes, mission-style roadmap unlocking, etc.) is being explored to increase engagement and long-term retention, especially among school-age users.

1) LIMITATION AND JUSTIFICATION FOR LLM BASELINES

Although Large Language Models (LLMs) such as GPT and PaLM are increasingly prevalent in natural language tasks,

a direct performance baseline was not conducted for several critical reasons.

1) **Computational Cost and Resource Constraints:** Running a large-scale reproducible evaluation of proprietary LLMs (e.g., GPT-4 or Gemini) requires significant financial commitment and dedicated computational resources that fall outside the scope of this academic research prototype.
2) **Lack of Deterministic Explainability:** The primary contribution of Sankalp is its explainable, verifiable reasoning via the Knowledge Graph and Rule Engine. LLMs operate as probabilistic "black boxes," making their outputs inherently nonauditable and unsuitable for high-stakes career guidance requiring guaranteed constraint adherence (e.g., minimum academic prerequisites).
3) **Ethical Compliance (IRB):** Reliance on external, nonauditable LLM APIs for sensitive user profiling and recommendation could introduce unquantifiable bias and create data compliance challenges (IRB/GDPR/IT Act) that contradict our commitment to responsible AI.

Our architectural comparison (Table 2) substantiates why a dedicated hybrid system provides superior rigor, efficiency ($\approx$ 350 ms core latency), and constraint adherence over general-purpose LLMs.

These enhancements not only address the current system constraints but also align with the broader goals of scalability, inclusiveness, and deeper personalization. As AI-driven guidance systems become central to educational planning, Sankalp Positions itself as a forward-compatible, contextually grounded, and socially responsible solution.

VIII. FUTURE WORK

- **Formal Expert Validation and External Validity:** To transition above the internal validation using the Curated Ground Truth (CGT), future work will focus on conducting a formal large-scale A/B evaluation with certified external career counselors. This validation will incorporate statistical measures such as **Cohen's Kappa** (κ) to quantify the inter-rater reliability between the system's Top-N recommendations and the consensus of the expert panel, thus strengthening the external validity of the system.
- **Advanced Adaptability and Generative Reasoning:** Sankalp's adaptive learning capabilities will be extended by deploying the RL model in a live environment to capture more diverse user feedback. In addition, integration with advanced generative models such as GPT for deeper causal reasoning and nuanced career storytelling will be explored to further personalize the guidance experience.
- **Mobile Deployment and Scalability:** The system architecture is primed for deployment as a mobile application to increase reach and accessibility. Future work includes packaging the front-end (React) and optimizing the back-end infrastructure for massive scalability, particularly focusing on efficient load balancing for the Naukri API requests and cloud-based voice processing.

IX. REPRODUCIBILITY

To promote transparency and ensure full reproducibility of the reported results, the core components of the Sankalp framework will be released to the academic community. Upon acceptance for publication, the complete source code will be made available in an open repository. This repository will contain the multi-agent orchestration logic, the implementation of the hybrid recommendation engine (including the SBERT module, the Rule-Based Engine, and the Reinforcement Learning fusion layer), the anonymized Curated Ground Truth (CGT) dataset used for validation, and the schema for the Knowledge Graph. This commitment allows for independent validation and future expansion of the methodologies developed in this research.

X. ETHICS AND RESPONSIBLE AI

The deployment of AI in sensitive domains, such as career counseling, requires a strong commitment to ethical principles, fairness, and transparency. Sankalp has been designed with these responsibilities in mind, ensuring that its recommendations are inclusive, explainable, and aligned with the broader values of social good and equal opportunity.

A. FAIRNESS AND BIAS MITIGATION

Sankalp uses multilingual SBERT embeddings and interpretable rule-based logic to minimize algorithmic bias. Unlike opaque black-box systems, Sankalp's explainable knowledge graph and transparent scoring mechanism allow users to understand why a particular career was suggested. By supporting regional languages (Hindi, Kannada, English) and taking into account diverse academic, emotional, and personality inputs, it reduces urban or English language biases that often exclude underserved communities.

B. ETHICS, PRIVACY, AND COMPLIANCE

All participants provided their informed consent prior to participating in the Sankalp system and participating in the evaluation study. Data collected during experiments were treated with strict adherence to privacy and security protocols. The user profile data was securely stored in a PostgreSQL database utilizing AES-256 encryption for data-at-rest protection. No personally identifiable information was shared with third parties. Data retention was strictly limited to the duration of the study, after which all records were discarded. The design of Sankalp ensures compliance with established regional and international data protection regulations, including GDPR principles and the Information Technology Act of India, ensuring user confidentiality and upholding ethical research practices in all phases of deployment and evaluation.

C. TRANSPARENCY AND USER AUTONOMY

Sankalp provides users with the ability to accept, reject, or ask for alternative career suggestions. This reinforces user agency in the decision-making process. Furthermore, all scoring and recommendation pathways are auditable and deterministic, supported by a rule engine that prioritizes explainability.

By combining responsible AI principles with robust architecture, Sankalp aims to set a standard for ethical, scalable, and user-centered AI systems in education and career development.

XI. CONCLUSION

This paper presented **Sankalp**, an AI-powered, multilingual, and emotionally aware career guidance system tailored for the diverse and large-scale educational landscape of India. By integrating deterministic rule-based reasoning with dynamic semantic understanding (via SBERT) and enhancing adaptability through a Reinforcement Learning (RL) layer, Sankalp offers a highly personalized and accessible guidance experience. Its modular, multi-agent design supports real-time voice/text input, full multilingual support (English, Hindi, Kannada), and explainable AI through a visual Knowledge Graph.

The performance of the system, validated against a Curated Ground Truth (CGT) dataset, demonstrated significant statistical superiority over baseline models, achieving a robust **Top − 3** Hit Rate of 90.5% (95% CI: [88.1%, 92.9%]). Operational analysis confirmed a highly efficient core logic, contributing a mean latency of **350** ms to the overall end-to-end response. These results validate the strength of the hybrid design in both precision and usability. The high User Satisfaction Score (4.71/5.0) further confirms the practical utility and positive reception of Sankalp's empathetic responses (VADER) and inclusive framework.

The value of Sankalp goes beyond the technological novelty. It aligns with the objectives of India's National Education Policy (NEP) 2020 and the Sustainable Development Goals (SDG 4 & 8) by providing a scalable and intelligent solution that addresses educational inequities and prepares youth for a dynamic workforce.

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APPENDIX

REPOSITORY AND CODE ACCESS

Upon acceptance, the complete Sankalp source code, including data preprocessing scripts, model configuration files, and the anonymized Curated Ground Truth (CGT) dataset schema, will be hosted in an open-access GitHub repository (link will be activated after publication).

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